

Camilo Laiton

Machine Learning & Computer Vision Engineer — Deep Learning for Large-Scale Image Data

Seattle, WA | [Portfolio](#) | [GitHub](#) | [LinkedIn](#)

SUMMARY

Machine learning and computer vision engineer specializing in deep learning and cloud-scale image processing. Built and operate petabyte-scale pipelines that have processed 3+ PB of image data while cutting costs by 67% per dataset and deliver production-ready deep learning models.

EXPERIENCE

Allen Institute for Neural Dynamics | Seattle, WA

10/2023 – Present

Software Engineer III — Machine Learning & Computer Vision

- Processed **3+ Petabytes** of light-sheet data through a cloud pipeline, cutting per-dataset cost from **\$150+ to \$50 on average (67% reduction)**. Results under review for publication.
- Built a **light-sheet brain template and cross-validated registration pipeline** to the Allen CCF v3 using ANTs, improving structure overlap from 0.70 to 0.85 Dice (**20% higher accuracy**) and cutting mean edge error from 53 μm to 24 μm (**55% lower**; $p < 0.001$ vs. direct registration), with more consistent results across brain structures.
- Parallelized template creation across cloud instances, cutting build time from **30 hours to 5 hours per template (6x speedup)** and making large cohorts (e.g., 80 brains) tractable where sequential processing would take 16 days.
- Built an **MAE-pretrained deep learning model for protein prediction** (anti-Bassoon / anti-PSD95) from the NHS-ester pan-protein channel, enabling synaptic marker inference without dedicated stains. Trained with a combined Dice + Focal loss; **achieved 14% lower validation loss than a traditional U-Net baseline** with visibly fewer false positives and over-segmentation artifacts.
- Developing a **multi-resolution multi-modal foundational image encoder** for brain light-sheet microscopy to support downstream tasks (segmentation, detection, classification) via self-supervised pretraining. Pretrained on 3 petabytes of lightsheet image data and evaluated on multiple downstream tasks.
- Re-engineered Cellpose to run at scale on cloud resources for large-scale cell segmentation, cutting end-to-end runtime from **46 hours to 4 hours (12x speedup)** on a 200 GB dataset by optimizing GPU gradient prediction, gradient/probability combination, and mask computation.
- Built a framework for fast, large-scale cell detection (ZYX positions) in light-sheet-acquired brains, plus large-scale mRNA identification using cloud compute and retrospective flat-field correction.
- Authored Nextflow pipelines and deployed Apptainer/Singularity (OCI) images for image processing in the cloud and on SLURM clusters.
- Mentored an undergraduate engineer (subsequently admitted to a Johns Hopkins CS PhD) and co-supervised a master's student, co-authoring two publications including a Best Paper Award at the International Conference on Artificial Intelligence Society 2026.
- Led code reviews and partnered with scientists to define software requirements; production code released open-source through the Allen Institute for Neural Dynamics.

Allen Institute for Neural Dynamics | Seattle, WA

07/2022 – 10/2023

Software Engineer II — Computer Vision

- Built image processing pipelines for light-sheet microscopy on AWS with Nextflow: destriping, 3D reconstruction, OME-Zarr conversion, atlas registration, and exploratory cell/mRNA analysis.
- Developed alignment methods that reduced median cell-coordinate error from **10.7 μm to 1.7 μm (85% reduction)**, below single-voxel resolution and consistent across datasets.
- Built a bioimage analysis package to estimate stitching errors in microscopy datasets and validated a destriping method that fully recovered the artifact-free baseline (Line Difference Metric reduction from **0.70 to 0.63**, PVS reduction from **157.19 to 153.05**) on simulated stripes; handled TB-scale data transfer from high-resolution microscopes via HPC.

Institución Universitaria ITSA | Barranquilla, Colombia

04/2021 – 07/2022

Software Engineer

- Surveyed state-of-the-art tools for brain anatomical structure segmentation and built a Python module to orchestrate them automatically on Linux.

- Built simulation software using evolutionary algorithms to reduce time, emissions, and cost across multimodal transport routes (river/rail/road) for Colombia's ferrofluvial 4.0 project; analyzed dual-use electromobility technologies.

- Developed a non-immersive virtual reality web platform to optimize time and resources for Ladys Confecciones S.A.S.

- Implemented a geoscientific data lake and ETL processes for the Servicio Geológico Colombiano (SGC), with performance tuning on mappings, sessions, and workflows to raise throughput.

TECHNICAL SKILLS

Machine Learning: PyTorch, TensorFlow/Keras, 3D segmentation & detection, image registration, self-supervised / foundation models, multi-task models

Cloud & Scale: AWS (Batch, S3), Azure, Dask, Zarr / OME-Zarr, TensorStore, Nextflow, Apptainer/Singularity, SLURM, HPC

Domain: Light-sheet & EM microscopy, atlas registration (Allen CCF v3), cell segmentation/detection, mRNA identification, petabyte-scale pipeline engineering

Core: Python, deep learning, large-scale data processing, pipeline design/deployment/optimization, ETL

Languages: English (MET C1), Spanish, French (DELF B2)

EDUCATION

M.Sc. in Computer Science, National University of Colombia (2020 – 2022) – Thesis: [Method for the segmentation of brain magnetic resonance images using a neural network architecture based on attention models.](#)

B.Sc. in Systems Engineering, University of Magdalena (2016 – 2020) – Thesis: [A CNN-based model for dynamic traffic signal timing estimation at simple urban intersections.](#)

CERTIFICATIONS

- DeepLearning.AI TensorFlow Developer Specialization
- Creating Multi-Task Models with Keras
- MET C1 (English)
- DELF B2 (French)

SELECTED PUBLICATIONS

- [Topographic structure and function of locus coeruleus norepinephrine neurons.](#) **Nature** (2026). *Contributing author (data processing).*
- [3D-MAESTRO: A scalable, modular, portable pipeline for automated processing of large-scale volumetric brain microscopy data.](#) **Under review** in eLife (2026). *Co-first author.*
- [Radon-Guided Wavelet-Domain Attention U-Net for Periodic Artifact Suppression in Brain MRI.](#) *Last author (mentor).*
- Methodology for Generation of a Synthetic Artifact Dataset in Magnetic Resonance Imaging of the Brain — **Best Paper Award**, International Conference on Artificial Intelligence Society 2026 (Valencia, Spain). *Last author (mentor).*
- [SmartSPIM Pipeline: A Scalable Cloud-Based Image Processing Pipeline for Light-Sheet Microscopy Data](#) *First author.*
- [Cloud Pipelines for Large-Scale Lightsheet Image Processing](#) *Second author.*
- [Deep 3D Neural Network for Brain Structures Segmentation Using Self-Attention Modules in MRI Images](#) *First author.*
- [Industry 4.0 Technologies Applied to the Rail Transportation Industry: A Systematic Review](#) *First author.*
- [A CNN-Based Model for Dynamic Traffic Signal Timing Estimation at Simple Urban Intersections](#) *First author.*